

AMAT113 Calculus II

Homework 4

Trigonometric Substitutions, Partial Fractions, and Improper Integrals

Submission Instructions

- Homework is due Sunday at 11:59 PM Eastern Time.
- Please print this PDF and complete the problems neatly by hand whenever possible.
- If printing is not available, completing the work on loose-leaf paper is acceptable.
- Show all necessary work. Answers without supporting work may receive reduced credit.
- Scan all completed pages, combine them into one PDF file, and submit the PDF through Brightspace before the deadline.
- Before submitting, confirm that every page is included, upright, legible, and in the correct order.

Part I: Trigonometric Substitutions

The following trigonometric substitutions are used to handle integrals involving radicals:

$$\sqrt{a^2 + x^2} \quad \text{requires} \quad x = a \tan \theta$$

$$\sqrt{a^2 - x^2} \quad \text{requires} \quad x = a \sin \theta$$

$$\sqrt{x^2 - a^2} \quad \text{requires} \quad x = a \sec \theta$$

1. Compute the following integrals:

(a) $\int \frac{1}{\sqrt{x^2 + 1}} dx$

(b) $\int \sqrt{x^2 + 4} dx$

Part I: Trigonometric Substitutions (continued)

2. Compute the following integrals:

(a) $\int \frac{1}{x^2 \sqrt{9 - x^2}} dx$

(b) $\int \frac{1}{x^2 \sqrt{x^2 - 4}} dx$

(c) $\int_{-1}^2 \sqrt{4 - x^2} dx.$

Part II: Partial Fractions

1. Compute the following integrals:

(a) $\int \frac{1}{x(x-1)(x-2)} dx$

(b) $\int \frac{4x-7}{x^2-x-6} dx$

Part II: Partial Fractions (continued)

2. Compute the following integrals:

(a) $\int \frac{x+2}{(x-1)^2} dx$

(b) $\int \frac{x+7}{x^2(x-2)^2} dx$

Part III: Improper Integrals

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

If the limit above exists, we say that the improper integral **converges**. Otherwise, it is said **diverge**.

1. Determine whether the following improper integrals converge or diverge. If the integral converges, then compute its value.

(a) $\int_1^\infty \frac{1}{x^3} dx$

(b) $\int_1^\infty \frac{x}{\sqrt{x^2 + 1}} dx$

(c) $\int_{-\infty}^2 e^{2x} dx$

Part III: Improper Integrals (continued)

The improper integral $\int_{-\infty}^{\infty} f(x) dx$ is defined by

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^{\infty} f(x) dx.$$

The integral on the left is said to converge only when **both** integrals on the right converge. Otherwise it is said to diverge.

2. Determine whether the following improper integrals converge or diverge. If the integral converges, then compute its value.

(a) $\int_{-\infty}^{\infty} \frac{1}{4+x^2} dx$

(b) $\int_{-\infty}^{\infty} \frac{x}{1+x^2} dx$

Part III: Improper Integrals (continued)

Comparison Test: Suppose f and g are continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$. Then

(a) If $\int_a^\infty g(x) dx$ converges, so does $\int_a^\infty f(x) dx$

(b) If $\int_a^\infty f(x) dx$ diverges, so does $\int_a^\infty g(x) dx$

3. Use the Comparison Test to determine whether the improper integral converges or diverges.

(a) $\int_1^\infty e^{-(1+x^2)} dx$

(b) $\int_1^\infty \frac{1}{\sqrt{1+x^5}} dx$

(c) $\int_1^\infty \frac{e^x}{\sqrt{1+x^2}} dx$

(d) $\int_1^\infty \frac{1}{\sqrt{e^{x^2} + x + \cos(x)}} dx$

End of Homework 4